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HO CHI MINH UNIVERSITY OF TECHNOLOGY

FACULTY OF CHEMICAL ENGINEERING



MECHANICAL PROCESSING

Laboratory Of Unit Operations (CH3015)

GRINDING – SIEVING – MIXING

CLASS CC03 — GROUP 9 — SEMESTER 252

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Ho Chi Minh City, March 2026

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1 Objectives

- Crushing a material; based on the sieve results, determine the size distribution of the material after grinding, the power consumption, and the productivity of the mill.
- Sieve the material after crushing; determine the sieve efficiency, build the cumulative distribution curve of the material after grinding, and then determine the material size after grinding.
- Mix two materials to determine the mixing index at several time intervals; build the mixing index graph over time to determine the appropriate mixing time.

2 Theory

2.1 Capacity and Performance of the Mill

The efficiency calculations of the mill use input particle size D_{p1} (ft) and the size of 80% of product after milling D_{p2} (ft).

P is capacity for milling material from a very large size to D_p (volume/time, with $i = \infty$):

$$P = K_b \sqrt{\frac{1}{D_p}} \quad (1)$$

According to the definition, W_i is the necessary energy to mill from a very large size down to $100 \mu\text{m}$ (kWh/ton). The relation between W_i and K_b (the Bond coefficient, which depends on the type of material and milling machine) is:

$$60W_i = K_b \sqrt{\frac{1}{100 \times 10^{-3}}} \Rightarrow K_b = \frac{60W_i}{\sqrt{10}} \approx 19W_i \Rightarrow P = 19W_i \frac{1}{\sqrt{D_p}} \quad (2)$$

where $P_1 = 19W_i/\sqrt{D_{p1}}$ and $P_2 = 19W_i/\sqrt{D_{p2}}$.

Milling capacity for crushing 1 ton of material over 1 minute from D_{p1} to D_{p2} :

$$P = P_2 - P_1 = 19W_i \left(\frac{1}{\sqrt{D_{p2}}} - \frac{1}{\sqrt{D_{p1}}} \right) \quad (3)$$

T is productivity (ton/min). Capacity for milling T tons/min from D_{p1} to D_{p2} :

$$P = 19W_i \left(\frac{1}{\sqrt{D_{p2}}} - \frac{1}{\sqrt{D_{p1}}} \right) T \quad (\text{kW}) \quad (4)$$

where D_{p1} , D_{p2} are the sizes of the material and product in mm. If P is multiplied by $4/3$, the consumption capacity of the motor is:

$$P = UI \cos \varphi \quad (5)$$

Where: U – voltage (V); I – amperage (A); $\cos \varphi$ – power factor.

Milling efficiency:

$$H = \frac{P}{P'} \times 100\% \quad (6)$$

2.2 Size Distribution

$$\frac{-d\Phi}{dD_p} = K D_p^b \quad (7)$$

Where: Φ – cumulative mass fraction on D_p ; D_p – particle size; K , b – coefficients describing particle distribution.

Integrating from $\Phi = \Phi_1$ to $\Phi = \Phi_2$ corresponding to $D_p = D_1$ and $D_p = D_2$:

$$\Phi_2 - \Phi_1 = \frac{K}{b+1} (D_{p1}^{b+1} - D_{p2}^{b+1}) \quad (8)$$

Between the n -th and $(n+1)$ -th sieve, using standard sieves where $D_{pn-1}/D_{pn} = r = \text{const}$:

$$\Delta\Phi_n = \Phi_n - \Phi_{n-1} = \frac{-K}{b+1} (D_{pn}^{b+1} - D_{pn-1}^{b+1}) \quad (9)$$

Replacing $D_{pn-1} = r \cdot D_{pn}$:

$$\Delta\Phi'_n = \frac{K(r^{b+1} - 1)}{b+1} D_{pn}^{b+1} = K' D_{pn}^{b+1} \quad \text{with} \quad K' = \frac{K(r^{b+1} - 1)}{b+1} \quad (10)$$

or equivalently:

$$\log \Delta\Phi_n = (b+1) \log D_{pn} + \log K' \quad (11)$$

K' and b are determined by plotting $\Delta\Phi_n$ vs. D_{pn} on a log-log graph.

2.3 Sieve Performance

$$E = \frac{J}{Fa} \times 100 \quad (12)$$

Where: F – initial material mass input to the sieve (g); J – material mass passing through the sieve (g); a – particle fraction that can pass through the sieve (%).

$F \cdot a$ is determined in the laboratory by sifting F to find J_1 , then sifting the remainder $F - J_1$ to find J_2 , and so on. The total $J_1 + J_2 + J_3 + \dots$ will asymptote to $F \cdot a$. Sieve performance is 100% if $J_1 = F \cdot a$.

2.4 Mix Equation

When mixing amount a of component A with amount b of component B to create a homogeneous mixture, the composition in an ideal mixture is:

- With A: $C_A = \frac{a}{a+b}$
- With B: $C_B = \frac{b}{a+b}$

These components would be equal in any volume of the mixture in an ideal case. In reality, since mixing time cannot be infinite, components A and B will differ across different volume elements. To evaluate homogeneity, we consider the mean standard deviation:

$$s_A = \sqrt{\frac{\sum_{i=1}^N (C_A - C_{iA})^2}{N-1}} \quad s_B = \sqrt{\frac{\sum_{i=1}^N (C_B - C_{iB})^2}{N-1}} \quad (13)$$

The smaller s_A , s_B are, the more ideal the mixture. To evaluate the homogeneity in a more practical way, the mixing index is:

$$I_s = \frac{\sigma_e}{s} \quad (14)$$

with σ_e the standard deviation of an ideal mixture:

$$\sigma_e = \sqrt{\frac{C_A C_B}{n}} \quad (15)$$

Therefore:

$$I_s = \sqrt{\frac{C_A C_B (N - 1)}{n \cdot \sum_{i=1}^N (C_A - C_{iA})^2}} \quad (16)$$

where n is the number of particles in the case of loose material mixing.

3 Experimental Method

3.1 Mill

- Turn on the machine without load to measure the no-load power.
- Give 200 g of rice to the mill.
- Turn on the input screw and press the meter to measure the maximum load current intensity.
- When the current returns to the no-load value, press the meter to determine the grinding time.

3.2 Sieve

- Determine sieve efficiency at the 0.2 mm sieve size: sieve 80 g of ground rice 5 times, each time for 5 minutes, and weigh the rice passing through the sieve each time.
- Determine the size distribution of the material after grinding: sieve 80 g of ground rice through various sieves of different sizes for 20 minutes and weigh the accumulated rice in each sieve.

3.3 Mix

- Put 1.5 kg of green beans and 3 kg of soya beans into the machine.
- Turn on the machine.
- Stop the machine at 6 time intervals (5 s, 15 s, 30 s, 60 s, 120 s, 300 s) and take 8 samples at each time according to the sampling chart. Count each particle in the sample.

The 8 sample positions are arranged in the mixing vessel as follows:

1	2	3
	4	5
6	7	8

4 Experimental Results

4.1 Results

4.1.1 Mill

Table 1: Grinding experiment raw data

Mass	Time	Amperage (A)	
		No Load	Load
200 g	26.92 s	3	3.8

Table 2: Size distribution after milling

Sieve Size (mm)	Mass (g)	Mass Distribution	Mass Percentage
0.425	49.47	49.47	0.618375
0.315	5.771	55.18	0.68975
0.200	14.35	69.53	0.869125
0.160	4.47	74.00	0.925

Table 3: Logarithmic values for size distribution

$\log(D_p)$	$\log(\Delta\Phi)$	Φ
-0.5	-0.2087	0.618375
-0.6	-1.1465	0.68975
-0.8	-0.7462	0.869125
-1.0	-1.2528	0.925

Power consumption for the motor of the loaded mill:

$$P_{\text{load}} = UI \cos \varphi = 220 \times 3 \times 0.8 = 528 \text{ W}$$

Power consumption for the motor of the no-load mill:

$$P_{\text{no-load}} = UI \cos \varphi = 220 \times 3.8 \times 0.8 = 668.8 \text{ W}$$

Power consumption for the material grinding engine:

$$P' = P_{\text{load}} - P_{\text{no-load}} = 668.8 - 528 = 140.8 \text{ W}$$

4.1.2 Sieve

Table 4: Sieve efficiency experiment (0.2 mm sieve)

No. of times	Time (min)	Mass distribution (g)	$\sum J_i$ (g)
1	5	25.44	25.44
2	5	0.43	25.87
3	5	0.24	26.11
4	5	0.08	26.19
5	5	0.02	26.21

4.1.2.1 Sieve efficiency at 0.2 mm:

Table 5: Size distribution of ground material

Sieve Size (mm)	Mass (g)
0.425	49.47
0.315	5.71
0.200	14.35
0.160	4.47

4.1.2.2 Size distribution after grinding:

4.1.3 Mixing

Raw count data (S = Soya beans, G = Green beans):

Table 6: Raw mixing data: particle counts per sample per time

Sample	5 s		15 s		30 s		60 s		120 s		300 s	
	S	G	S	G	S	G	S	G	S	G	S	G
1	99	184	83	111	81	115	64	67	63	171	67	65
2	75	114	82	58	114	89	79	65	104	59	70	96
3	125	84	132	57	98	107	69	43	98	113	131	114
4	144	79	122	73	119	98	109	108	128	97	134	108
5	151	99	127	76	144	127	122	69	104	63	135	86
6	141	93	141	181	109	142	107	160	113	150	133	152
7	98	148	90	148	98	108	108	145	104	108	118	176
8	112	100	96	156	59	119	98	156	118	71	112	146

Calculated mixing indices at each time step ($C_A = 2/3$, $C_B = 1/3$):

Table 7: Calculated mixing index I_s at each time

Time	$\sum$	N	σ_e	s	I_s
5 s	0.27751	1846	0.05510	–	–
15 s	0.28518	1733	0.05610	–	–
30 s	0.34596	1727	0.05102	–	–
60 s	0.28310	1569	0.05918	–	–
120 s	0.30013	1664	0.05581	–	–
300 s	0.28308	1843	0.05460	–	–

4.2 Graphs

4.2.1 Grinding

- Chart 1: Chart of $\sum J_i$ following sifting time.
- Chart 2: Chart of $\log \Delta \Phi_n$ following $\log D_{pn}$.

4.2.2 Sifting

- Chart 3: Chart of accumulated size distribution of the material on sieve.

4.2.3 Mixing

- Chart 4: Chart of mixing index I_s following time.

4.3 Discussion

4.3.1 Adaptation of Bond's Law

Based on the milling theories, we can see that:

- **Rittinger's Surficial Theory:** Applicable only when the energy provided per unit weight of solid is not too large. It has limited practical value due to complex determination of K_r for each material-machine combination.

- **Kick's Volumetric Theory:** Based on stress analysis in the elastic limit. This theory also lacks high practical value due to its complicated coefficient determination.
- **Bond's Law:** This is the highest practical-value theory for milling efficiency prediction. The work index W_i includes the internal friction inside the milling machine. It has small error when applied to different milling machines of the same type and for both dry and wet milling processes. Therefore, this theory is very convenient for calculations.

4.3.2 Comments on Milling and Sifting Efficiency

Milling: $E = 19.44\%$

The milling efficiency is calculated from:

- The weight of material milled M (by scale).
- Milling time (by stopwatch).
- The current at maximum load measured by ammeter.

The efficiency can be considered to be relatively low due to:

- *Objective reasons:* The inherently low efficiency of the grinding machine.
- *Subjective reasons:* Weighing errors and imprecise time adjustment. However, these cause very minor errors that do not significantly affect the result.
- *Measurement after sifting:* Because the material particles are very light and small, they easily disperse into the surrounding environment. Additionally, some material remains stuck on the surface of the sieve due to particle size. This has the most significant effect on result error.

Sifting: $E = 97.01\%$

The sifting efficiency is high, thanks to:

- Low humidity in the material.
- Suitable thickness of the material layer on the sieve, allowing particles to pass through faster and more easily.
- The flat surface of the sieves.

4.3.3 Reliability of Results

- **Milling:** Reliability is low. The most effective factors are described above in the efficiency discussion.
- **Sifting:** Reliability is high. Key factors include low material humidity, suitable layer thickness, flat sieve surfaces, and simple calculations that minimise computational error.
- **Mixing:** Reliability is pretty high. Factors affecting the mixing result:
 - *Bean size distribution:* The different sizes between soya beans and green beans affect the mixing process.
 - *Mixing time:* Measured manually by stopwatch, so there will be small timing errors (considered negligible).
 - *Specific weight:* Green beans and soya beans have approximate specific weights, which supports the mixing process.
 - *Breakability:* These beans are not easily breakable, making the process easier to operate.
 - *Sample positions:* Samples are taken from the arrangement corresponding to the defined diagram, ensuring spatial uniqueness and improving result accuracy.
 - *Simple calculation:* Minimises calculation-induced error.

4.3.4 Comment on Sampling Method in Mixing Experiment

Samples are taken at 6 different times: 5 s, 15 s, 30 s, 60 s, 120 s, and 300 s. At each time, 8 samples are taken at positions distributed across the mixing vessel. We must take samples at those positions to ensure analysis of all bean types, making each sample spatially unique so that results have higher accuracy. Because during the mixing process, not all positions have the same bean distribution, we take from many positions and calculate the average.

Additionally, we take samples at 6 different times to analyse the change of mixing index over time, allowing us to find the optimal mixing time at which the highest uniformity is achieved.

4.3.5 Reliability of Mixing Results

The reliability of the mixing result is pretty high. Elements affecting the result significantly:

- **Bean size distribution:** Different sizes between green beans and soya beans.
- **Mixing time:** Determined manually by stopwatch; error is negligible.
- **Specific weight:** Green beans and soya beans have approximate weights, supporting the process.
- **Breakability:** These beans are not breakable, making operation easier.
- **Sample positions:** Taken from defined positions in the vessel, ensuring spatial uniqueness.
- **Simple calculation:** Result calculation is simple so the error is too small to mention.

5 Appendix

5.1 Milling Calculation

5.1.1 Determination of the Equivalent Diameter of the Material Particle

The equivalent diameter is the diameter of a sphere having the same ratio V/S as the real particle. For a spherical particle, $D_{p,eq} = D$. For a cylinder with length L and diameter D :

The average dimensions of the rice:

- Length: $L = 6$ mm
- Diameter: $D = 1.5$ mm

Considering the rice as a cylinder:

$$\frac{V}{S} = \frac{\frac{\pi D^2 L}{4}}{\pi D L + 2 \cdot \frac{\pi D^2}{4}} = \frac{D \cdot L}{4L + 2D}$$
$$D_{p,eq} = \frac{4V}{S} = \frac{4DL}{4L + 2D} = \frac{4 \times 1.5 \times 6}{4 \times 6 + 2 \times 1.5} = 2 \text{ mm}$$

Therefore: $D_{p1} = D_{p,eq} = 2$ mm.

5.1.2 Chart $\log \Delta \Phi_n$ following $\log D_{pn}$

Line equation from the theory:

$$\log \Delta \Phi_n = (b + 1) \log D_{pn} + \log K'$$

According to Chart 2, the diagram passes through 3 points, giving:

$$b + 1 = 2.9975 \Rightarrow b = 1.9975 \quad \log K' = 2.4921 \Rightarrow K' = 310.53$$

The sieve ratio:

$$r = \frac{D_{pn-1}}{D_{pn}} = \frac{1}{3} \left(\frac{0.425}{0.315} + \frac{0.315}{0.2} + \frac{0.2}{0.16} \right) = 1.39$$

Therefore:

$$K = \frac{K'(b+1)}{r^{b+1} - 1} = \frac{310.53 \times 2.9975}{1.39^{2.9975} - 1} = 552.93$$

5.1.3 Size Distribution of Material on Sieve

The differential equation:

$$\frac{-d\Phi}{dD_p} = K D_p^b \Rightarrow \Phi = \frac{-K}{b+1} D_p^{b+1} + C = -103.6 D_p^{2.9975} + C$$

When $D_p = 0.425$: $\Phi = 0.39 \Rightarrow C = 8.36$.

$$D_p = \sqrt[2.9975]{\frac{\Phi - 8.36}{-103.6}}$$

By Bond's Law, D_{p2} is the size such that 80% of product weight passes through the sieve ($\Phi = 0.20$):

$$D_{p2} = 0.43 \text{ mm}$$

5.1.4 Milling Efficiency Calculation

Theoretical milling capacity (dry process, $W_i = 13 \text{ kWh/ton}$):

$$T = \frac{M}{t} = \frac{200 \times 10^{-6} \text{ ton}}{26.92/60 \text{ min}}$$

$$P = \frac{4}{3} \times 19 \times 13 \times \left(\frac{1}{\sqrt{0.43}} - \frac{1}{\sqrt{2}} \right) \times T = 0.13 \text{ kW}$$

Milling machine power:

$$P' = U \cdot I \cdot \cos \varphi = 220 \times 3.8 \times 0.8 = 668.8 \text{ W} = 0.6688 \text{ kW}$$

Milling efficiency:

$$H = \frac{P}{P'} \times 100\% = \frac{0.13}{0.6688} \times 100\% = 19.44\% \quad (17)$$

5.2 Sifting Experiment Calculation

5.2.1 Determination of $F \cdot a$

Based on Chart 1, the $\sum J_i$ curve asymptotes to:

$$F \cdot a = 25.77 \text{ g}$$

5.2.2 Sifting Efficiency

$$E = \frac{J_1}{F \cdot a} \times 100\% = \frac{25}{25.77} \times 100\% = 97.01\% \quad (18)$$

5.3 Mixing Experiment Calculation

The constituents of Soya beans (A) and Green beans (B) in an ideal mixture are:

$$C_A = \frac{a}{a+b} = \frac{3}{3+1.5} = 0.67 \quad C_B = 1 - C_A = 0.33$$

The standard deviation of an ideal mixture:

$$\sigma_e = \sqrt{\frac{C_A C_B}{n}}$$

where $N = 8$ (number of samples) and n is the number of beans per sample volume.

The mixing index for each time step:

$$I_s = \sqrt{\frac{C_A C_B (N-1)}{n \cdot \sum_{i=1}^N (C_A - C_{iA})^2}}$$